

HOPE, BC, Canada

WMO: 711870

Lat: 49.3699N

Lon: 121.4935W

Elev: 128

StdP: 14.63

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.5	19.3	-0.5	5.4	18.3	5.5	7.3	21.6	23.2	33.5	21.5	31.8	16.0	110	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	87.1	67.4	83.5	66.2	80.2	65.0	69.3	83.2	67.5	80.7	65.8	78.0	9.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	89.6	75.6	62.3	84.4	73.0	60.8	80.0	70.9	33.5	83.6	31.9	81.0	30.6	78.4	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.3	16.4	DB	11.5	94.1	5.7	4.1	7.4	97.1	4.1	99.5	0.9	101.8	-3.3	104.8	(4)
(5)			WB	9.2	72.2	5.4	2.3	5.4	73.8	2.2	75.1	-0.8	76.4	-4.7	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	35.6	39.1	43.6	49.9	56.9	61.2	66.7	67.0	61.4	51.0	41.5	35.4	(6)
(7)		DBStd	12.70	7.52	5.95	5.68	5.35	5.99	5.57	5.38	4.94	5.61	4.87	6.43	6.72	(7)
(8)		HDD50	1788	446	305	208	64	9	0	0	0	0	45	257	454	(8)
(9)		HDD65	5423	911	725	663	452	260	143	40	35	138	433	704	919	(9)
(10)		CDD50	2096	0	0	10	62	223	335	517	527	341	77	4	0	(10)
(11)		CDD65	256	0	0	0	1	9	28	92	97	29	0	0	0	(11)
(12)		CDH74	2640	0	0	1	18	147	336	977	881	279	1	0	0	(12)
(13)		CDH80	774	0	0	0	2	29	101	315	258	69	0	0	0	(13)
(14)	Wind	WSAvg	7.6	9.3	8.4	7.3	7.0	7.0	7.2	7.5	7.1	6.6	6.6	7.7	9.1	(14)
(15)	Precipitation	PrecAvg	61.6	8.6	5.3	6.7	4.0	3.1	3.1	1.5	1.9	3.3	6.9	9.5	8.1	(15)
(16)		PrecMax	98.1	14.5	12.2	13.0	8.1	7.0	5.9	2.9	8.1	7.4	16.4	27.3	13.2	(16)
(17)		PrecMin	48.0	5.0	0.3	0.7	1.5	0.8	0.0	0.1	0.0	0.0	1.3	2.8	4.1	(17)
(18)		PrecStd	10.2	2.8	2.9	3.2	1.7	1.4	1.2	0.8	1.8	2.0	3.6	4.9	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	54.7	66.9	76.3	83.8	88.8	93.4	90.7	86.3	70.1	56.5	50.5	(19)
MCWB			47.4	46.2	51.9	58.5	62.5	67.9	69.1	66.8	64.8	57.7	52.9	46.8	(20)	
2%		DB	48.4	51.2	59.9	69.8	78.1	82.7	87.3	86.2	81.2	64.5	53.5	47.2	(21)	
		MCWB	45.1	45.0	49.1	55.0	61.2	65.1	68.8	67.4	63.5	54.8	50.4	44.5	(22)	
5%		DB	45.6	48.8	55.4	64.8	73.5	77.3	83.1	82.6	76.6	60.8	51.1	44.7	(23)	
		MCWB	42.8	44.0	47.0	52.3	59.6	63.2	67.2	66.6	62.8	54.1	48.6	42.2	(24)	
10%		DB	43.6	46.6	51.8	60.1	68.8	72.8	79.3	79.1	72.1	57.9	49.2	42.9	(25)	
		MCWB	41.2	42.8	45.6	50.2	57.3	60.8	65.6	65.4	60.6	53.0	47.0	40.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	49.3	53.6	59.9	64.7	69.2	73.0	70.9	67.6	60.7	54.3	47.9	(27)
MCD B			50.7	51.2	62.7	73.7	79.6	85.5	88.5	84.7	81.3	66.0	56.1	49.5	(28)	
2%		WB	45.6	46.9	50.6	56.0	62.3	66.0	70.0	69.0	65.4	57.5	51.2	45.0	(29)	
		MCD B	47.7	49.2	57.3	66.8	75.2	80.1	84.3	82.3	77.4	61.2	52.8	46.5	(30)	
5%		WB	43.5	45.0	48.4	53.4	60.2	63.6	67.9	67.3	63.5	55.4	49.3	42.9	(31)	
		MCD B	45.1	47.8	53.5	62.7	71.4	75.8	80.9	80.2	74.0	59.0	50.7	44.1	(32)	
10%		WB	41.8	43.2	46.6	51.1	58.0	61.3	66.0	65.6	61.5	53.6	47.5	41.1	(33)	
		MCD B	43.3	46.0	50.9	58.2	67.5	71.2	78.1	77.2	70.0	57.0	48.8	42.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	10.8	13.7	16.6	17.4	16.0	18.9	18.9	17.3	11.9	7.8	6.6	(35)
MCDBR			9.7	13.9	21.2	26.2	26.7	26.4	26.4	26.5	26.1	17.1	9.5	8.6	(36)	
MCWBR			7.3	8.8	11.8	13.9	13.4	13.1	12.2	11.8	12.4	9.8	7.3	6.8	(37)	
5% WB		MCD BR	9.3	11.6	17.4	22.9	23.5	24.2	24.2	23.6	23.1	13.8	9.0	8.5	(38)	
		MCWBR	7.6	8.2	10.6	12.6	12.3	12.4	11.8	11.2	11.5	8.8	7.5	7.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.299	0.316	0.361	0.399	0.377	0.365	0.370	0.353	0.365	0.315	0.293	(40)	
taud		2.306	2.332	2.310	2.246	2.282	2.388	2.443	2.456	2.487	2.387	2.442	2.344	(41)		
Ebn at Noon		236	269	284	280	273	280	282	275	269	240	232	222	(42)		
Edh at Noon		22	27	33	40	40	36	34	32	28	26	20	19	(43)		
(44)	All-Sky Solar Radiation	RadAvg	336	610	954	1431	1687	1777	2053	1751	1217	699	346	259	(44)	
(45)		RadStd	43	59	98	100	146	183	183	125	107	79	37	33	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+58
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page